Project Assignment 4

Compiler Design Lab
Prof. Dr. Sebastian Hack
Fabian Ritter, B.Sc.

Project Task F Intermediate Representation

- Install LLVM 15.0. Follow the instructions from the "Introduction to LLVM" slide set to do so, preferably using the `build_llvm.sh` script from the project template.

  Building LLVM will take a while.

- Download and study the provided example programs to gain some intuition about the LLVM API. The examples show how to directly construct LLVM IR representation for small but relevant code fragments. You should be able to build and run them if you add the binary folder of your LLVM install (using the defaults from our build script, this is `llvm/install/bin`) to your path.

- Implement a systematic LLVM IR construction from your AST. The compiler must perform this if `--compile` or no explicit compilation mode is given. Use LLVM's 8-bit and 32-bit integer types to implement the `char` and `int` types in C, respectively. You may use LLVM's `llvm::DataLayout` to compute sizes of types.

- The output shall be human readable LLVM code into a file. When the input is `foo/bar.c` the output file is `bar.ll` in the current directory. To output into a file perform the following:

```
#include "llvm/Support/FileSystem.h"
#include "llvm/Support/raw_ostream.h"
...
std::error_code EC;
raw_fd_ostream stream(filename, EC, llvm::sys::fs::OpenFlags::OF_Text);
M.print(stream, nullptr); /* M is a llvm::Module */
```

- The soft deadline for this milestone is 2023-01-27.

- Keep it simple!